HAROON **RAZA**

1 x Instrument Engineer | 1 x Trainee Engineer (Embedded Systems) | 4 x Intern (VoIP Engineering, Embedded Systems, Engineering and General)

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Professional Summary

Electrical engineer with diverse hands-on experience across industrial asset integrity, IT automation and embedded systems. Currently an Instrument Engineer at VELOSI, building ISO-based Asset Registers, verifying equipment against P&IDs and performing Equipment Criticality Analysis (ECA). Previously automated VoIP infrastructure with Docker, CI/CD and AWS at GOMARKHO, and built ESP32-based IoT and automation systems through internships and personal projects.

Experience

Instrument Engineer, *VELOSI, Islamabad* Mar 2026 – Present

- Develop and maintain the Asset Register (AR) by extracting technical data from site photos and documents and tagging components on site images.
- Build ISO-based equipment hierarchies to structure plant assets in the AR.
- Break equipment down into ISO-based subunits to support detailed asset records.
- Identify the fluid service of equipment and lines from P&IDs and site data.
- Verify components listed in the AR against P&IDs and their physical location on site.
- Inspect site equipment to assess its condition and support asset integrity.
- Perform Equipment Criticality Analysis (ECA) to assess and rank equipment by its criticality to operations.

Trainee Engineer, Embedded Systems, *CMGL, Islamabad (Remote)* Jul 2025 – Dec 2025

- Designed an ESP32-based LoRa GPS tracker for real-time location monitoring in remote, low-connectivity areas.
- Developed a two-tank water level controller using conductive and ultrasonic sensing for redundancy.
- Built an IoT CNC machine using Arduino UNO for motion control and ESP32-CAM for remote operation with live video.
- Created a universal IR repeater that records and replays commands, controlled remotely over MQTT.
- Implemented persistent storage with ESP32 NVS and SPIFFS to retain settings and calibration across reboots.

VoIP Engineering Intern, *GOMARKHO, Islamabad* Feb 2025 – Apr 2025

- Implemented call routing, interception and session management in FreeSWITCH for business PBX use.
- Containerised VoIP services with Docker and automated deployment using CI/CD pipelines on AWS.
- Automated server setup and disk monitoring with Bash scripts.

Embedded Systems Intern, *Emertxe, Bengaluru (Remote)* Jan 2025 – Feb 2025

- Built a PIC16F877A microwave simulator with keypad input and LCD feedback.
- Applied debouncing and interrupt handling to ensure reliable user input.

Engineering Intern, *National Institute of Electronics, Islamabad* Jun 2024 – Aug 2024

- Rebuilt the driver library of a dot matrix display to restore and control its output.
- Contributed YOLO-based vehicle detection and EasyOCR license plate recognition to a smart parking prototype.

Operations Manager, *IEEE COMSATS University Islamabad* Feb 2024 – Jun 2024

- Introduced G Suite tools to coordinate tasks across IEEE events.

Intern, *Nobel Learning PBC, USA (Remote)* Jul 2023 – Dec 2023

- Wrote technical content for the “Data Clustering & Visualization” course module.

Projects

Smart Cart ML Module *(Final Year Project)* Raspberry Pi 5, PyTorch, LangChain, ChromaDB

- Developed a ResNet18 few-shot object detector with barcode fallback that learns new products from fewer than five samples each.
- Benchmarked the detector at 88.4% accuracy and 9.2 FPS on a Raspberry Pi 5, outperforming a YOLO11 baseline (68.4%).
- Built a RAG chatbot for product queries using LangChain, ChromaDB and Ollama.

Pylex *(open source)* Python, PaddleOCR, ONNX Runtime

- Developed a Python OCR toolkit that converts scans, images and PDFs into searchable PDFs without altering the original page.
- Handled rotated pages and dense technical drawings using overlapping tile processing and targeted vertical-text retry.
- Packaged the toolkit as a batch CLI with parallel workers and a reusable Python API.
- Used ONNX Runtime to cut benchmark processing time from 10.3 s to under 1 s per page.

Multi-Source Home Automation System *(Real Time Embedded Systems final project)* ESP32, Firebase, MIT App Inventor

- Built an ESP32 controller that prioritises solar, then grid, then generator, and starts the generator within seconds of a dual-source failure.
- Added CT-based load monitoring across all three power sources.
- Developed a Firebase-connected mobile app with live data, fault alerts and remote control of 4+ appliances.

Other Projects

- **PID Line Following Robot (Robian, inter-university competition):** Built an auto-calibrating six-sensor robot with PID control and sharp-turn handling.
- **Generator Auto On/Off:** Automated generator startup within seconds of grid loss, with a fault indicator.
- **deen-cli (open source):** Developed a Python terminal app for prayer tracking and Quran browsing with private local records in SQLite.

Skills

Industrial	Asset Integrity, Asset Register Development, ISO-based Equipment Hierarchy, Fluid Service Identification, P&ID Verification, Equipment Criticality Analysis (ECA)
Embedded	ESP32, Arduino, PIC, Raspberry Pi, LoRa, MQTT, Circuit Simulation (Proteus), PCB Design (Altium)
Programming	Python, C, C++, Bash, MATLAB, SQL (SQLite)
IT and Cloud	Linux, Git, Docker, CI/CD, AWS, FreeSWITCH, Firebase
AI and Data	PyTorch, OpenCV, YOLO, OCR (PaddleOCR, EasyOCR), RAG, Data Analysis
Languages	English (Full Professional), Urdu (Native), Punjabi (Native)

Education

BSc Electrical Engineering, *COMSATS University Islamabad, CGPA 3.73/4.00* Sep 2021 – Jul 2025

- Final Year Project: Smart Cart ML Module (see Projects).
- Coursework (Instrumentation and Control): Electrical Measurements and Instrumentation, Control Systems, Digital Signal Processing, Signals and Systems.
- Coursework (Embedded and Computing): Real Time Embedded Systems, Microprocessor Systems and Interfacing, Artificial Intelligence, Neural Networks, Data Communication and Computer Networks.
- Led lab groups through hands-on engineering coursework.
- Volunteered on the security team at PSYWSC'23 (IEEE).

Certifications

Applied Data Science Lab (WorldQuant University); AI Agents Course (Hugging Face); MATLAB Programming for Engineers and Scientists (Coursera); Python Essentials I and Programming Essentials in C++ (Cisco); Python Beyond the Basics (10Pearls); 30 Days of LeetCode Challenge (GDSC COMSATS).

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https://voltvirtuoso.github.io/assets/media/Haroon-Raza-Resume_d0734376.pdf

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